BioTwin AI — Real-Time T2D Digital Human Twin Architecture

Happiest Health Digital Twin Challenge 2026 | Team Apex (IIT Jodhpur)

SECURE SANDBOX BOUNDARY (DPDP ACT 2023 & HIPAA COMPLIANT | 100% SYNTHETIC)

1. DATA LAYER (Synthea & ODE)

- Synthea EHR (1,000 pts)
  HbA1c, Lipids, eGFR, BP
  Indian BMI Cutoffs, PRS
- Bergman Minimal ODE
  Continuous Glucose (5m)
  Gastric Emptying Lag
- Indian Meal Library
  48 Regional Dishes
  Carbs & Glycemic Index

2. FEATURE STREAM (Batch & Streaming)

- Shared Windows
  6h, 24h, 7d lookbacks
  Zero Train-Serve Skew
- Real-Time Ring Buffers
  O(1) Streaming Ingest
  Gaps & Outlier Handling
- Metabolic Features
  TIR, TAR, Glucose Slopes
  HRV, Sleep Stages, Steps

3. ML FUSION ENGINE (Trained XGBoost)

- XGBoost Fusion Model
  Static + Dynamic Concatenation
  AUROC: 0.998, AUPRC: 0.999
- Isotonic Calibration
  Brier Score: 0.0012
  Empirical Reliability
- Quantile Forecaster
  +30, +60, +90, +120m
  Coverage Calibrated
- SHAP Explainer (XAI)
  Real-Time Feature Drivers
  Clinical Natural Language

4. API BACKEND (FastAPI & WebSockets)

- Real-Time Stream
  WebSocket & SSE Fan-out
  60x Replay Engine
- Alert Fatigue Engine
  6 Alerts/Day Cap, Deduplication
  Intervention Snooze
- What-If Simulator
  Indian Diet Perturbation
  Counterfactual Finder
- Clinical Exports
  HL7 FHIR R4 Bundle
  PDF Consult Report

5. DOCTOR DASHBOARD (Next.js 14)

- Live CGM Visualizer
  Target Bands (70-180)
  Projected Spikes
- What-If Sliders
  Roti, Biryani, Walks
  Risk Delta Reduction
- Anatomical Twin
  3D Organ Indicators
  5-Year Fast Forward
- Clinician Copilot
  NVIDIA NIM LLM
  Automated Action Dispatch

Built for Happiest Health Digital Twin Challenge 2026 | Full Source Available at github.com/Mayank-iitj/biotwin-ai